

Rohit Yadav

Built 25+ LLMs from scratch • 120M–1.4B params • TPU clusters • Trained on 500 Billion+ tokens

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EDUCATION

National Institute of Technology, Jalandhar

Bachelor of Technology in Information Technology

Jalandhar, India

Aug 2023 – Aug 2027

Green Valley School for Children

CBSE Board 12th

Gandhinagar, India

May 2021 – May 2023

EXPERIENCE

AI Research Intern — Molsys Pvt. Ltd.

Jun 2026 – Present

AI/ML Research Internship (Rs. 75,000/- Stipend) — Remote

- Developed and optimized **LLM pipelines** spanning pretraining, supervised fine-tuning (SFT), model distillation, RLHF-inspired alignment workflows, and efficient inference.
- Built **agentic AI systems** with intelligent model routing, tool use, and orchestration capabilities while exploring cost-performance optimization using Amazon Bedrock.
- Researched **distributed training, large-scale inference, quantization, and system-level optimization**, enabling deployment of foundation-scale models under practical compute constraints.

Research Lead — AutonomousX

Apr 2026 – Present

Open Source LLM Research Organization

- Received **Google TPU Research Cloud (TRC)** support for large-scale LLM training and research.
- Trained **25+ LLMs (120M–1.4B parameters)** on **500B+ tokens** using TPU pods ranging from v4-8 to v4-32 and v6e-64.

Open Source Contributor

March, 2026 – Present

Meta LLaMA and Google DeepMind Gemma

- Contributed to production-scale LLM repositories by fixing critical bugs and improving usability.
- Google DeepMind Gemma** — resolved tokenizer normalization and jax typing issues.
- Meta LLaMA** — fixed incorrect CLI import causing ModuleNotFoundError in download pipeline.

Full-Stack Web Developer Intern

Jun 2025 – Jul 2025

Skylark Express Delhi Pvt Ltd

On-site, Gurugram

- Developed internal logistics dashboard improving data visibility and workflow efficiency.
- Improved UI responsiveness and system performance for internal operations.

PROJECTS

AutonomousX: TPU-Native LLM Training Organization (Instinct Model Family)

[\[HuggingFace\]](#)

AI/ML • LLMs • Transformers • Distributed Systems

- Built and open-sourced **25+ LLMs (120M–1.4B params)** by engineering **TPU-native distributed pipelines (JAX pmap)** under AutonomousX.
- Trained the **Instinct model family** on **500B+ tokens** (PILE, DOLMA) under extreme overtraining regimes to study **scaling laws, convergence, and degeneration dynamics**.
- Open-sourced complete pipelines including HuggingFace weights, training logs, and a 10-minute **TPU setup guide** enabling full reproducibility for researchers.
- Provided a **ready-to-run inference notebook** allowing quick evaluation and usage of the trained models.

PathoPreter: Clinical-Grade SNV Pathogenicity Ranker (DNA Foundation Model)

[\[HuggingFace\]](#)

AI/ML • GenAI for Biology • Foundation Models • Bioinformatics

- Developed a **hybrid genomic foundation model (500M params)** fusing a Nucleotide Transformer backbone with clinical tabular features for **SNV pathogenicity prediction**.
- Achieved **0.9186 ROC-AUC** on a rigorous 14k rare-variant benchmark, outperforming industry SOTA (AlphaMissense, CADD) by **+0.32 ROC-AUC**.

- Designed a **ranking-first clinical triage system** demonstrating **75%+ pathogen recall** while testing only 5–10% of variants, drastically reducing lab turnaround time and costs.
- Validated biological learning via **SHAP ablation studies** (64.9% reliance on raw DNA) and open-sourced the **end-to-end reproducible pipeline**.

Data Science Project: Large-Scale Genomic Dataset Pipeline (1.25M Variants)

[\[GitHub\]](#)

Data Science • Data Engineering • ML Infrastructure • Big Data

- Architected an end-to-end **32-core parallelized data pipeline**, reducing processing time for 2.5TB+ genomic datasets from 150+ hours to under 20 hours.
- Transformed **7.7M raw ClinVar records** into a leakage-free dataset of **1.25M high-quality SNVs** optimized for ML evaluation and benchmarking.
- Integrated annotations from **ClinVar, gnomAD, dbNSFP, and hg38** to generate 50+ predictive features alongside GRCh38 DNA sequence mappings.
- Implemented strict preprocessing to remove **HGVS identifiers and data leakage**, created a balanced 300K-variant subset, and engineered **GPU-optimized NumPy (.npy) shards** for accelerated transformer training and data loading.

BroLang Compiler + Custom Virtual Machine (Java & JVM-style Execution Model)

[\[GitHub\]](#)

Systems Programming • Compilers • Runtime Systems • C++

- Designed a complete compiler from scratch for a custom language (BroLang), implementing a **Lexer, Parser, AST, and Bytecode Generator**.
- Built a **custom 16-bit Virtual Machine (VCPU + 65KB RAM)** to execute compiled bytecode, inspired by **Java + JVM-style execution**.
- Developed a full runtime system supporting **instruction decoding, stack-based execution, control flow, and I/O operations**, enabling high-level programs to run on custom virtual hardware.
- **Tech Stack:** Modern C++17, custom ISA design, bytecode architecture, manual memory management.

OS-Level AI Agent: Autonomous System Controller via LLM Tool Execution

[\[GitHub\]](#)

AI Agents • LLM Systems • Tool Use • Automation

- Built an **OS-level AI agent** that converts any LLM API into a **fully autonomous system controller** capable of executing real-world tasks.
- Designed a **planner → action → feedback loop** with **self-healing execution**, enabling multi-step reasoning and autonomous recovery from failures such as missing dependencies.
- Implemented system tools including **shell command execution, file I/O, web interaction, and Python runtime**, enabling full OS control.
- Integrated optional browser automation (Playwright) and built a **GUI-based control interface** for real-time task execution.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript/JavaScript, SQL

AI & Distributed Training: LLM Pretraining & Fine-tuning, TPU (v4/v6e) & GPU (A100/H100) Clusters, JAX **pmap**, Scaling Laws, Mixed Precision (BF16), XAI (SHAP)

Frameworks & Web: JAX/Flax, PyTorch, HuggingFace, React, Next.js, FastAPI, PostgreSQL, MongoDB

Cloud & Systems Architecture: GCP (TPU Research Cloud), AWS, Docker, Linux, Compiler Design, Virtual Machines, POSIX Sockets

ACHIEVEMENTS

Awarded Google TRC grant for 320 TPU access (SPOT + STANDARD VMs).

Awarded AMD MI300X GPU access for 300 non-preemptive hours

IOQM (PRMO): Qualified for Indian Olympiad Qualifier in Mathematics

[\[Certificate\]](#)

KVPY SA: AIR 2590

[\[Certificate\]](#)

LeetCode: Knight (2013), 600+ problems solved across all platforms

[\[Profile\]](#)

CodeChef: 4-Star (1818)

[\[Profile\]](#)

Codeforces: Specialist (1417)

[\[Profile\]](#)

CodeChef START166B: Global Rank 9

[\[Contest\]](#)